Straightforward calculations finally lead to the following solution for the Bloch vector $\hat{\mathbf{O}}$

$$\begin{pmatrix}\hat{\sigma}_x(\nu)\\ \hat{\sigma}_y(\nu)\\ \hat{\sigma}_z(\nu)\end{pmatrix} = \begin{pmatrix}A_{xx}(\nu,t') & A_{xy}(\nu,t') & A_{xz}(\nu,t')\\ A_{yx}(\nu,t') & A_{yy}(\nu,t') & A_{yz}(\nu,t')\\ A_{zx}(\nu,t') & A_{zy}(\nu,t') & A_{zz}(\nu,t')\end{pmatrix}\begin{pmatrix}\hat{\sigma}_y(0)\\ \hat{\sigma}_y(0)\\ \hat{\sigma}_z(0)\end{pmatrix} + \begin{pmatrix}A_{x0}(\nu,t')\hat{\mathbb{I}}\\ A_{y0}(\nu,t')\hat{\mathbb{I}}\\ A_{z0}(\nu,t')\hat{\mathbb{I}}\end{pmatrix}, \tag{D11}$$

where

$$\begin{aligned}
A_{xx}(\nu,t) &= \frac{1}{2}e^{-2\gamma_t\nu}\left[2e^{\gamma_t\nu}\cos^2[\theta_t]\cos[\nu\varepsilon_t] - \cos[2\theta_t] + 1\right]\\
A_{xy}(\nu,t) &= A_{yx}(\nu,t') = e^{-\gamma_t\nu}\cos[\theta_t]\sin[\nu\varepsilon_t]\\
A_{xz}(\nu,t) &= A_{zx}(\nu,t') = \frac{1}{2}e^{-2\gamma_t\nu}\sin[2\theta_t]\left[1 - e^{\gamma_t\nu}\cos[\nu\varepsilon_t]\right]\\
A_{yy}(\nu,t) &= e^{-\gamma_t\nu}\cos[\nu\varepsilon_t]\\
A_{zz}(\nu,t) &= \frac{1}{2}e^{-2\gamma_t\nu}\left[2e^{\gamma_t\nu}\sin^2[\theta_t]\cos[\nu\varepsilon_t] + \cos[2\theta_t] + 1\right]
\end{aligned} \tag{D12}$$

with γ_t given by Eq. (D5). The explicit (rather cumbersome) expressions for the functions $A_{j0}(\nu,t')$ $(j = x,y,z)$ will not be given here, since they drop out of the final expressions due to the fact that $[\![\mathrm{cov}_t^u(.,\hat{\mathbb{I}})]\!] = 0$.

3. Excess heat cumulant generating function

We are now ready to compute the cumulant generating function of the dissipated heat in the slow-driving regime, Eq. (D6). As stressed above, the calculations can be simplified by first noticing that $\mathrm{cov}_t^u(\cdot,\hat{\mathbb{I}}) = 0$; furthermore, it turns out that, for this model, we have that $\mathrm{cov}_t^u(\hat{\sigma}_y,\cdot) = 0$ and finally that $\mathrm{cov}_t^u(\hat{\sigma}_x,\hat{\sigma}_z) = \mathrm{cov}_t^u(\hat{\sigma}_z,\hat{\sigma}_x)$. Putting all this together, the result is given by

$$\begin{aligned}
\mathcal{K}_q(u) = -\epsilon\beta^2\int_0^1 dt\, &[\![\mathrm{cov}_t^u(\hat{\sigma}_z,\hat{\sigma}_z)]\!]\left[f_t^2\tilde{A}_{zz}(t) + f_t g_t\tilde{A}_{xz}(t)\right]\\
&+ [\![\mathrm{cov}_t^u(\hat{\sigma}_x,\hat{\sigma}_x)]\!]\left[g_t^2\tilde{A}_{xx}(t) + f_t g_t\tilde{A}_{zx}(t)\right]\\
&+ [\![\mathrm{cov}_t^u(\hat{\sigma}_z,\hat{\sigma}_x)]\!]\left[f_t^2\tilde{A}_{zx}(t) + g_t^2\tilde{A}_{xz}(t) + f_t g_t\left(\tilde{A}_{zz}(t) + \tilde{A}_{xx}(t)\right)\right],
\end{aligned} \tag{D13}$$

where $\tilde{A}_{jk}(t) \equiv \int_0^{+\infty} d\nu A_{jk}(\nu,t)$, which can be analytically computed. The above quantum covariances are given by

$$\begin{aligned}
[\![\mathrm{cov}_t^u(\hat{\sigma}_x,\hat{\sigma}_x)]\!] &= \frac{2u\varepsilon_t^2\sin^2[\theta_t](\beta-u) - 2\cos^2[\theta_t](-\cosh[\beta\varepsilon_t] + \cosh[\varepsilon_t(u-\beta)] + \cosh[u\varepsilon_t] - 1)}{\beta^2\varepsilon_t^2(\cosh[\beta\varepsilon_t] + 1)}\\
[\![\mathrm{cov}_t^u(\hat{\sigma}_x,\hat{\sigma}_z)]\!] &= \frac{\sin[2\theta_t]\mathrm{sech}^2\left[\frac{\beta\varepsilon_t}{2}\right]\left(-\cosh[\beta\varepsilon_t] + u\varepsilon_t^2(\beta-u) + \cosh[\varepsilon_t(u-\beta)] + \cosh[u\varepsilon_t] - 1\right)}{2\beta^2\varepsilon_t^2}\\
[\![\mathrm{cov}_t^u(\hat{\sigma}_z,\hat{\sigma}_z)]\!] &= \frac{2u\varepsilon_t^2\cos^2[\theta_t](\beta-u) - 2\sin^2[\theta_t](-\cosh[\beta\varepsilon_t] + \cosh[\varepsilon_t(u-\beta)] + \cosh[u\varepsilon_t] - 1)}{\beta^2\varepsilon_t^2(\cosh[\beta\varepsilon_t] + 1)}.
\end{aligned} \tag{D14}$$

Putting together all these results and substituting the functional expressions for ε_t and θ_t given below in Eq. (E2) allows to obtain a rather cumbersome but analytic expression for the integrand of the cumulant generating function $\dot{\mathcal{K}}_q(u,t')$, where $\mathcal{K}_q(u) \equiv \epsilon\int_0^1 dt'\dot{\mathcal{K}}_q(u,t')$. The final remaining integration over the rescaled time variable t however can be carried out only numerically.

4. The classical part of the cumulant generating function

Here we will provide the details of the calculations of the classical part of the cumulant generating function, in order to single out the quantum contributions due to coherences. Following the main theory outlined above and in the main text, we first need to compute the diagonal part of the power operator onto the instantaneous energy eigenbasis, i.e.

$$\dot{\hat{H}}_t^d = \mathcal{D}_{\hat{H}_t}\left(\dot{\hat{H}}_t\right) = \frac{1}{2}\dot{\varepsilon}_t\left(\cos[\theta_t]\hat{\sigma}_z + \sin[\theta_t]\hat{\sigma}_x\right). \tag{D15}$$

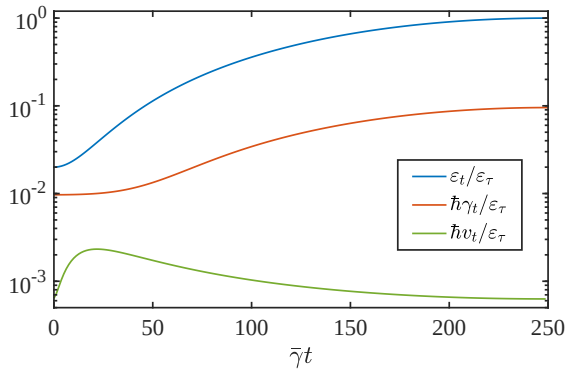

FIG. E1. Comparison between the energy splitting ε_t, the dissipation rate $\gamma_t = \alpha\varepsilon_t \coth(\beta\varepsilon_t/2)/2\hbar$ and the instantaneous driving speed $v_t = \sqrt{(\dot{\varepsilon}_t/\varepsilon_t)^2 + \dot{\theta}^2}$. The conditions for the secular and slow-driving approximations are seen to hold at all times. The parameters are the same as in Fig. 3, i.e. $\alpha = 0.191$, $\varepsilon_0 = 0.02\varepsilon_\tau$, $\beta\varepsilon_\tau = 20$ and $\bar{\gamma}\tau = 250$.

In order to then find $\dot{\hat{H}}_t^d(\nu)$ we can exploit the solutions Eq. (D11) for the evolved operators $\hat{\sigma}_z(\nu)$, $\hat{\sigma}_x(\nu)$. What thus remains to be calculated in order to determine $\mathcal{K}_q^d(u)$ defined in Eq. (A13) is the symmetrised covariance $\mathrm{cov}_t(\dot{\hat{H}}_t^d(\nu), \dot{\hat{H}}_t^d) = \frac{1}{2}\mathrm{Tr}\left(\{\dot{\hat{H}}_t^d(\nu), \dot{\hat{H}}_t^d\}\hat{\pi}_t\right) - \mathrm{Tr}\left(\dot{\hat{H}}_t^d(\nu)\hat{\pi}_t\right)\mathrm{Tr}\left(\dot{\hat{H}}_t^d\hat{\pi}_t\right)$. The calculations are simplified first by noticing that this quantity is symmetric in its two arguments and furthermore that $\mathrm{cov}_t(\cdot, \hat{\mathbb{I}}) = 0$ and $\mathrm{cov}_t(\cdot, \hat{\sigma}_y) = 0$. The remaining symmetric covariances are given by

$$\mathrm{cov}_t(\hat{\sigma}_x, \hat{\sigma}_x) = \frac{\left(e^{\beta\varepsilon_t} - 1\right)^2 \cos[2\theta_t] + 6e^{\beta\varepsilon_t} + e^{2\beta\varepsilon_t} + 1}{2\left(e^{\beta\varepsilon_t} + 1\right)^2} \tag{D16}$$

$$\mathrm{cov}_t(\hat{\sigma}_x, \hat{\sigma}_z) = -\sin[\theta_t]\cos[\theta_t]\tanh^2\left(\frac{\beta\varepsilon_t}{2}\right) \tag{D17}$$

$$\mathrm{cov}_t(\hat{\sigma}_z, \hat{\sigma}_z) = 1 - \cos^2[\theta_t]\tanh^2\left(\frac{\beta\varepsilon_t}{2}\right). \tag{D18}$$

A straightforward calculation finally shows that

$$\begin{aligned}\mathcal{K}_q^d(u) &= -\epsilon(u^2 - \beta u)\int_0^1 dt' \int_0^\infty d\nu \mathrm{cov}_t\left(\dot{\hat{H}}_t^d(\nu), \dot{\hat{H}}_t^d\right) \\ &= -\epsilon(u^2 - \beta u)\int_0^1 dt' \left(\frac{1}{2}\dot{\varepsilon}_t\right)^2 \Big[\mathrm{cov}_t\left(\hat{\sigma}_z, \hat{\sigma}_z\right)\cos[\theta_t]\left(\cos[\theta_t]\tilde{A}_{zz}(t) + \sin[\theta_t]\tilde{A}_{xz}(t)\right) \\ &\qquad + \mathrm{cov}_t\left(\hat{\sigma}_x, \hat{\sigma}_x\right)\sin[\theta_t]\left(\sin[\theta_t]\tilde{A}_{xx}(t) + \cos[\theta_t]\tilde{A}_{zx}(t)\right) \\ &\qquad + \mathrm{cov}_t\left(\hat{\sigma}_x, \sigma_z\right)\left(\tilde{A}_{zx}(t) + \sin[\theta_t]\cos[\theta_t]\left(\tilde{A}_{xx}(t) + \tilde{A}_{zz}(t)\right)\right)\Big].\end{aligned} \tag{D19}$$

E. Monte Carlo simulations and results

We now give further details on the Monte Carlo trajectory simulations used to obtain Fig. 3. In the following discussion we restore the non-rescaled time coordinate so that the protocol takes place in the time interval $t \in [0, \tau]$. We follow the standard quantum-jump approach [37] to unravel the Lindblad equation defined by Eq. (D3) into pure-state trajectories. The initial state $|\psi_0\rangle$ is randomly selected from the eigenstates of $\hat{H}_0$ according to the corresponding Boltzmann distribution. The time at which the first jump occurs is chosen by sampling from the waiting-time distribution $w_t = -(d/dt)\langle\psi_t|\psi_t\rangle$, where $|\psi_t\rangle$ obeys the non-Hermitian Schrödinger equation $(d/dt)|\psi_t\rangle = -i\hat{H}_t^{\mathrm{eff}}|\psi_t\rangle$ with initial condition $|\psi_0\rangle$, and the effective Hamiltonian is

$$\hat{H}_t^{\mathrm{eff}} = \hat{H}_t - \frac{i\alpha\varepsilon_t}{2}\left((N_t + 1)\hat{L}_t^\dagger\hat{L}_t + N_t\hat{L}_t\hat{L}_t^\dagger\right). \tag{E1}$$

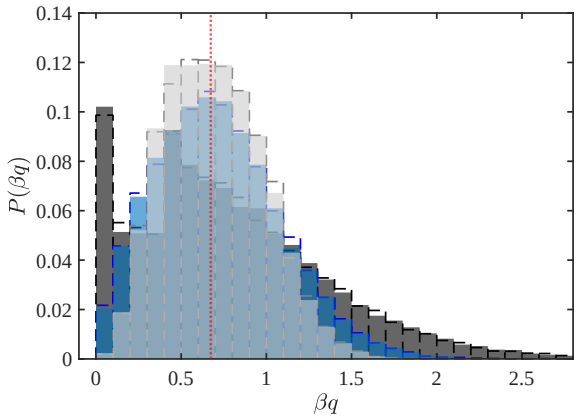

FIG. E2. Probability distribution of heat transferred during slow erasure protocols. The data correspond to classical (dashed lines) and quantum (solid bars) protocols of duration $\bar{\gamma}\tau = 100$ (dark grey), $\bar{\gamma}\tau = 250$ (blue) and $\bar{\gamma}\tau = 500$ (light grey). The Landauer limit $\beta q = -\Delta S$ is shown by the red dotted line. Other parameters are the same as in Fig. 3, i.e. $\alpha = 0.191$, $\varepsilon_0 = 0.02\varepsilon_\tau$, $\beta\varepsilon_\tau = 20$.

This evolution equation is solved efficiently using a fourth-order Runge-Kutta algorithm with adaptive time-step control [72]. If a jump occurs at time t, it corresponds either to emission, $|\psi_t\rangle \to \hat{L}_t |\psi_t\rangle$, or absorption, $|\psi_t\rangle \to \hat{L}_t^\dagger |\psi_t\rangle$, with probabilities given by the excited or ground state populations, respectively. The normalised post-jump state is then taken as a new initial condition and the procedure is repeated until the final time is reached. The heat distribution is constructed by simulating many such trajectories and recording the total quantity of energy transferred to the environment during each one, as in Eq. (2).

The specific protocol we consider is defined by

$$\varepsilon_t = \varepsilon_0 + (\varepsilon_\tau - \varepsilon_0)\sin^2(\pi t/2\tau), \qquad \theta_t = \pi(t/\tau - 1). \tag{E2}$$

Erasure corresponds to choosing the initial spectral gap to be far below the thermal energy, $\varepsilon_0 \ll k_B T$, and the final spectral gap to greatly exceed this energy, $\varepsilon_\tau \gg k_B T$. This ensures that the respective thermal states are effectively given by $\hat{\pi}_0 \simeq \hat{\mathbb{I}}/2$ and $\hat{\pi}_\tau \simeq |0\rangle\langle 0|$, respectively. The slow-driving regime corresponds to the case where the Hamiltonian changes slowly in comparison to the characteristic relaxation rate $\gamma_t = \frac{1}{2}\alpha\varepsilon_t \coth(\beta\varepsilon_t/2)$. The instantaneous driving speed can be quantified by the parameter $v_t = \sqrt{(\dot{\varepsilon}_t/\varepsilon_t)^2 + \dot{\theta}^2}$, which takes into account the rate of change of both the energy splitting ε_t and the mixing angle θ_t. In Fig. E1 we show that the conditions for the validity of both the slow-driving and secular approximations, namely $\hbar v_t \ll \hbar\gamma_t \ll \varepsilon_t$, hold at all times during the evolution for the parameters considered in this work.

In Fig. E2 we show the heat distributions obtained for various different values of τ, considering both quantum (where θ_t is given by Eq. (E2)) and classical (with $\theta_t = 0$) protocols. We find that the bulk of the distribution is very similar between the quantum and corresponding classical protocols. The only qualitative difference is seen in the extreme outliers which do not occur in classical protocols. Such outliers may correspond either to large heat transfer, $q \gg k_B T$, or, less frequently, a negative total heat transfer $q < 0$, which cannot occur classically. In both the classical and quantum case, the bulk of the distribution becomes increasingly concentrated around the Landauer bound and converges to a Gaussian shape as τ is increased.